

# HelloSupport — Design and development decisions

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**Final deliverable** required by the spec ([SPEC.md](#) §3). This document explains **every** design and development choice in the project.

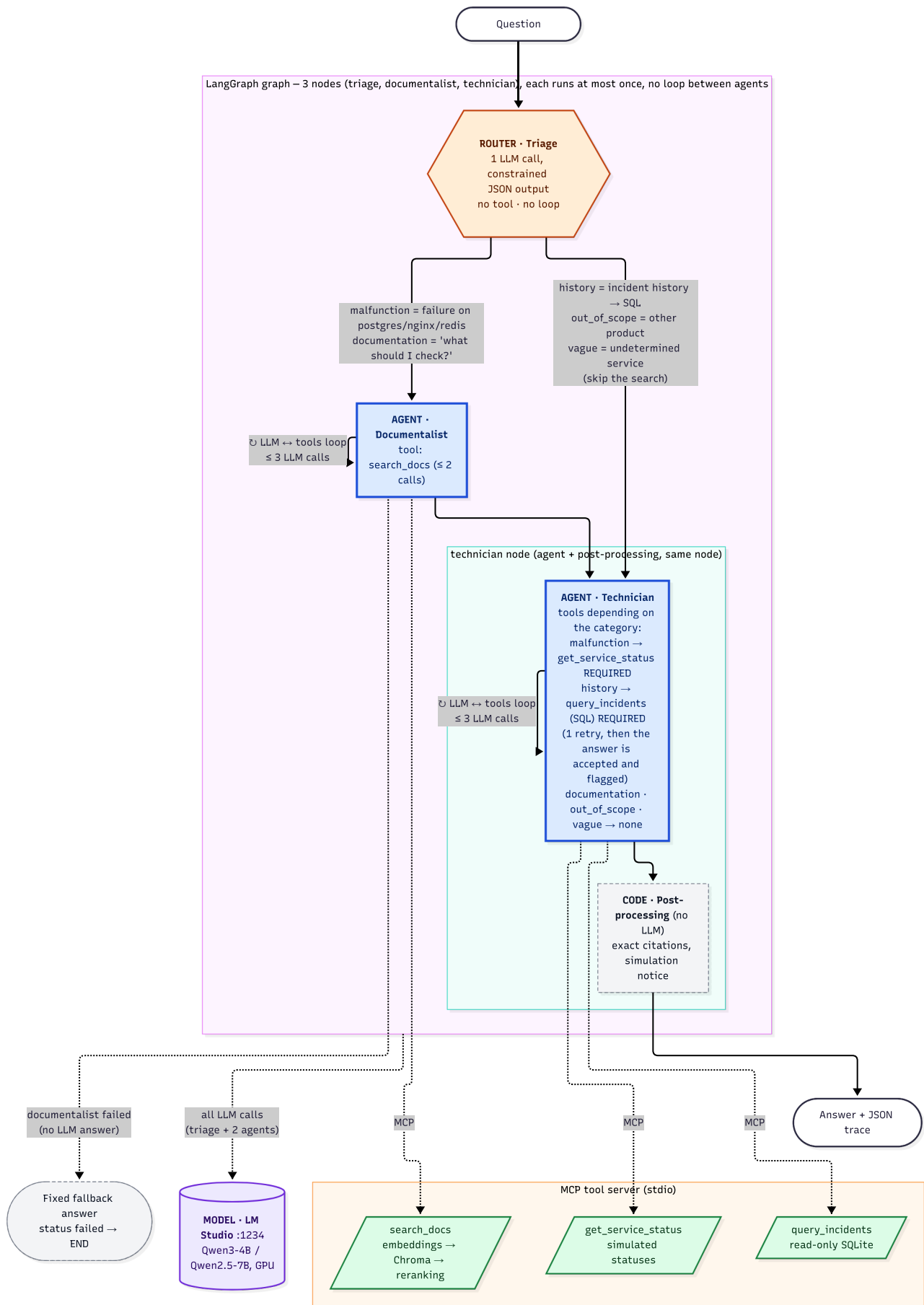
- **During the project:** ADR-style log. Each milestone adds its dated D-xx entries when the decision is made (with what was known at the time).
- **At the end (J5):** consolidation. Summary at the top, entries reviewed in light of the measurements ([BENCH.md](#)), PDF export ([DESIGN DECISIONS.pdf](#)). **Status: consolidated on 2026-10-02 (v1.8.0), 28 decisions; D-29 to D-31 added on 2026-10-04; aligned with the code of v1.11.2 on 2026-10-04 after the documentation-to-code review ([REVIEW DOC CODE.md](#)).**

Entry format: **Need** → **Options** → **Choice** → **Why** → **Trade-offs / limits** → **Skill demonstrated**. A revised decision is not deleted: it is marked Superseded by D-yy.

## Summary (consolidated on 2026-10-02 for v1.8.0, last updated on 2026-10-04 for v1.11.2)

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**What was built.** A troubleshooting assistant in the terminal and in the browser, 100% local. A question in French or English goes through **triage** (LLM, constrained JSON output), a **documentalist** (semantic search: embeddings → Chroma → reranking) and a **technician**. Depending on the intent, the technician observes a **simulated** service status, queries a read-only **SQL incident database** (text-to-SQL), or answers from the documentation. It cites its sources and separates what was observed from what is hypothetical. Tools go through an **MCP server**, orchestration through **LangGraph**, and the models (4B / 7B SLMs) are served by **LM Studio** on GPU.

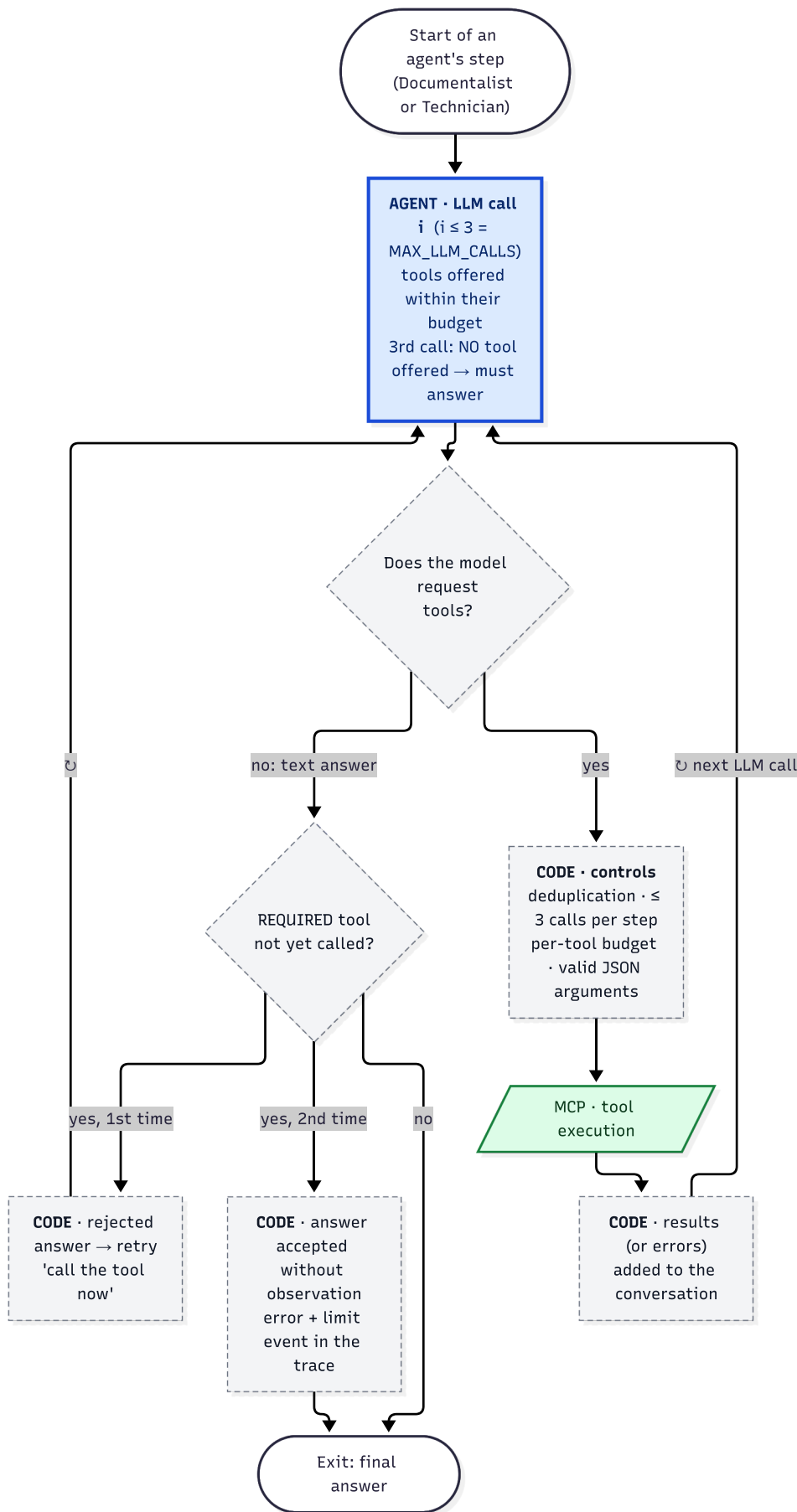


## Legend

| Style in the diagram                                                                                  | Kind            | What it is                                                                                                        | Loop?                                  |
|-------------------------------------------------------------------------------------------------------|-----------------|-------------------------------------------------------------------------------------------------------------------|----------------------------------------|
|  blue, thick border  | <b>AGENT</b>    | An LLM that chooses which tools to call and with which arguments, reads the results, then answers (run_agent)     | <b>yes:</b> LLM ↔ tools, ≤ 3 LLM calls |
|  orange hexagon      | <b>ROUTER</b>   | A single LLM call constrained by a JSON schema, which picks 1 of the 5 fixed categories. No tool, so not an agent | no                                     |
|  gray, dashed border | <b>CODE</b>     | Deterministic Python, no LLM                                                                                      | no                                     |
|  green parallelogram | <b>MCP TOOL</b> | Exposed by the separate MCP server, called by the agents                                                          | —                                      |
|  purple cylinder     | <b>MODEL</b>    | LLM served locally by LM Studio (OpenAI-compatible API)                                                           | —                                      |

**Agent, router or code?** We call an *agent* an LLM that **decides its own** actions (which tools, which arguments) and **loops** on their results until it can answer: this is the case for the Documentalist and the Technician. *Triage* is not an agent but a **router**. It makes a single LLM call, with no tool, whose output is constrained to {intent, service}; it is the **code** that derives the path and the allowed tools from it (D-17). *Post-processing* and the loop controls are deterministic **code** (D-18, D-22).

**Why no loop between agents?** The LangGraph graph is **linear**: triage → documentalist → technician for malfunction and documentation, triage → technician directly for history, out\_of\_scope and vague (D-30), and each node runs once per question. The only loops are **inside** each agent, and they are bounded (D-18, D-29):



- **At most 3 LLM calls** ( $MAX\_LLM\_CALLS$ ). The last one is made without tools: the agent always ends up answering in text.
- **At most 3 tool calls per step**, after deduplication. Each tool also has its own budget ( $search\_docs \leq 2$ ,  $get\_service\_status \leq 1$ ,  $query\_incidents \leq 2$ ).

- **Required tool:** for malfunction and history, a tool call is required. If the model answers without making one, the code rejects that answer and retries once. If it answers in text a second time, that answer is **accepted** (status stays done), but the code records the error "a tool call was required but never made; answer accepted without observation" and emits a `limit` event with the same text (D-17).
- **Exit:** the loop stops as soon as the model answers in text without requesting a tool.

### Mapping table component → decision → skill demonstrated

| Component             | Decision taken                                                   | Entries                | Skill demonstrated                     |
|-----------------------|------------------------------------------------------------------|------------------------|----------------------------------------|
| Model serving         | LM Studio, OpenAI-compatible API, GPU, Q4_K_M                    | D-03, D-04             | Model-serving platforms ; GPU          |
| Model choice          | 7B by default, 4B SLM as a measured option                       | D-04, D-25             | LLMs and SLMs                          |
| LLM integration       | openai SDK, wrapper that measures latency and tokens             | D-05                   | Model integration                      |
| Chunking + embeddings | ## sections, multilingual MiniLM (Sentence Transformers)         | D-06, D-07             | RAG, embeddings, Hugging Face, PyTorch |
| Vector database       | Persistent embedded Chroma, fingerprint-based index              | D-08                   | Vector search, vector databases        |
| Reranking             | Cross-encoder top-5 → top-3, off-topic threshold                 | D-09                   | Reranking                              |
| GPU                   | torch CUDA, warm-up                                              | D-10                   | GPU, latency                           |
| Tools                 | stdio MCP server, 3 typed tools, ToolError                       | D-11, D-12, D-15, D-20 | MCP, tool integration, tool calling    |
| Data                  | SQLite incident database, simulated scenarios                    | D-14                   | Database systems, SQL engines          |
| SQL tool              | Read-only text-to-SQL, 5 layers of protection                    | D-13, D-23             | Enterprise data agents                 |
| Orchestration         | LangGraph: triage → documentalist → technician, history branch   | D-16                   | Orchestration, LangGraph               |
| Tool decision         | Router with structured output + policy enforced by the code      | D-17, D-24             | AI agents, intelligent workflows       |
| Reliability           | Bounded loop, dedup, required checked, post-processing           | D-18, D-22             | Reliability, production-grade          |
| State and traces      | TypedDict + reducers, timestamped JSON trace                     | D-19                   | State management                       |
| Validation            | 6 executable cases, benchmark, deterministic checks              | D-21, D-28             | Prototype → production                 |
| Measurements          | Warm/cold latency, throughput at concurrency 1/4, estimated cost | D-26                   | Latency, throughput, cost              |
| Code                  | Single-responsibility modules, MCP boundary                      | D-02, D-27             | Software architecture, Python          |
| Framework             | 100% local, €0                                                   | D-01                   | Cost                                   |

### What the measurements taught us (the resulting decisions are marked "Decision driven by a measured failure"):

1. **Small models do not decide on their own to observe:** left free to choose, 0/3 for the 4B as for the 7B. A narrow router with constrained output, followed by a policy enforced by the code, gives 18/18 with the 7B (D-17).
2. **An API constraint is not a guarantee:** `tool_choice="required"` is not enforced by LM Studio, and the model invented a status. The code checks and retries (D-18). The final demo shows this guardrail in action.
3. **The model hallucinates when it "narrates" the next step** instead of doing it (C4). Having it emit all queries at once removes the problem (D-23).
4. **What the code knows, the code says:** simulated status, absence of action, list of valid sources (D-22).
5. **A prompt tweak must be measured on held-out data:** adding examples improved the SLM and degraded the 7B (D-24).
6. **Cost is driven by context, not by model size:** ~2,700 input tokens for ~370 output tokens (D-26).
7. **Automatic checks have bugs too:** two of them were wrong (D-21). Hence the archiving of full answers for review.

**Accepted limits:** 3 knowledge-base sheets, 14 incidents, 6 cases, one machine, temperature 0. No measurement is statistically sound. SQLite is not an enterprise engine. LM Studio is not a production server. When a tool fails, the

SLM can still invent sources, and the 7B asserted an unobserved status once in a 10-run control (D-25, D-30). Out of scope: distributed systems, real scale, leadership (see [SPEC.md](#) §7 and §10).

## Index

| #    | Decision                                                                                  | Milestone | Status                                           | Skill demonstrated                  |
|------|-------------------------------------------------------------------------------------------|-----------|--------------------------------------------------|-------------------------------------|
| D-01 | 100% local, no paid cloud API                                                             | scoping   | Accepted                                         | Cost, model serving                 |
| D-02 | Python 3.12 + uv + installable package with CLI                                           | J0        | Accepted                                         | Python, software architecture       |
| D-03 | LM Studio as the local serving platform                                                   | J0        | Accepted                                         | Model-serving platforms             |
| D-04 | Model choice: Qwen3-4B-Instruct-2507 (SLM) and Qwen2.5-7B-Instruct, Q4_K_M                | J0        | Accepted, confirmed by D-25                      | LLMs & SLMs, GPU                    |
| D-05 | LLM client: openai SDK via the OpenAI-compatible endpoint, thin in-house wrapper          | J0        | Accepted                                         | Model integration, latency/cost     |
| D-06 | Chunking knowledge-base sheets by ## section, doc_id#section citation                     | J1        | Accepted                                         | RAG, semantic retrieval             |
| D-07 | Embeddings: paraphrase-multilingual-MiniLM-L12-v2 (Sentence Transformers)                 | J1        | Accepted                                         | Embeddings, Hugging Face            |
| D-08 | Vector store: persistent embedded Chroma, index rebuilt on fingerprint change             | J1        | Accepted                                         | Vector search, vector databases     |
| D-09 | Reranking: cross-encoder mmarco-mMiniLMv2-L12-H384-v1, top-5 → top-3, off-topic threshold | J1        | Accepted                                         | Reranking                           |
| D-10 | Embedding/reranker inference on GPU (torch CUDA), warm-up at load time                    | J1        | Accepted                                         | GPU, PyTorch, latency               |
| D-11 | MCP: separate tool server, stdio, mcp SDK v2, verified with MCP Inspector                 | J2        | Accepted                                         | MCP, tool integration               |
| D-12 | Typed tool contracts (Literal → enum), explicit ToolError                                 | J2        | Accepted                                         | Tool calling, reliability           |
| D-13 | Read-only SQL tool: 5 layers (regex, LIMIT, mode=ro, authorizer, timeout)                 | J2        | Accepted                                         | Enterprise data agents, SQL engines |
| D-14 | Generated SQLite incident database (relative dates), simulated JSON status scenarios      | J2        | Accepted                                         | Database integration, data analysis |
| D-15 | Loading the RAG in a thread when the MCP server starts                                    | J2        | Accepted                                         | Latency                             |
| D-16 | LangGraph orchestration: triage → documentalist → technician, history branch              | J3        | Accepted                                         | Orchestration, LangGraph            |
| D-17 | Triage via structured output + tool policy enforced by the code                           | J3        | Accepted (Decision driven by a measured failure) | AI agents, intelligent workflows    |
| D-18 | Bounded agent loop: budgets, last call without tools, dedup, required checked             | J3        | Accepted                                         | Reliability, workflow execution     |
| D-19 | TypedDict shared state (reducers) + timestamped JSON trace with metrics                   | J3        | Accepted                                         | State management, latency/cost      |
| D-20 | Host MCP client with the mcp SDK directly (ToolBox), no LangChain adapter                 | J3        | Accepted                                         | MCP, tool integration               |
| D-21 | Validation: 6 executable cases, deterministic checks, bench with full answers             | J4        | Accepted                                         | Prototype → production, evaluation  |
| D-22 | Deterministic post-processing: normalized citations, simulation notice                    | J4        | Accepted (Decision driven by a measured failure) | Reliability                         |
| D-23 | Data questions: all SQL queries in the same step                                          | J4        | Accepted (Decision driven by a measured failure) | Enterprise data agents              |

| #    | Decision                                                                                   | Milestone | Status                                           | Skill demonstrated                         |
|------|--------------------------------------------------------------------------------------------|-----------|--------------------------------------------------|--------------------------------------------|
| D-24 | Triage: rules + examples, validated on held-out paraphrases (12/12)                        | J4        | Accepted (Decision driven by a measured failure) | Evaluate AI tech, SLMs                     |
| D-25 | <b>ADR:</b> default model = 7B (18/18); SLM faster but invents sources                     | J5        | Accepted                                         | LLMs & SLMs, latency/cost, evaluate → plan |
| D-26 | Measurement method: warm/cold latency, throughput at concurrency 1/4, estimated cost       | J5        | Accepted                                         | Latency, throughput, cost                  |
| D-27 | Code structure: single-responsibility modules, MCP boundary                                | J5        | Accepted                                         | Software architecture, Python              |
| D-28 | Testing strategy: offline unit tests, model integration tests, system benchmark            | J5        | Accepted                                         | Production-grade, reliability              |
| D-29 | Agents (LLM ↔ tools loop) vs router (triage) vs code; linear graph, no loop between agents | doc       | Accepted                                         | AI agents, orchestration                   |
| D-30 | out_of_scope and vague skip the documentalist (conditional edge), measured before/after    | v1.10.0   | Accepted                                         | Orchestration, latency/cost                |
| D-31 | Web demo: Starlette + SSE on the real pipeline, one question at a time, warm MCP server    | v1.11.0   | Accepted                                         | Production-grade, observability            |

## D-01 — 100% local, no paid cloud API

- **Date:** 2026-10-02 · **Milestone:** scoping (user decision)
- **Need:** practice LLMs, agents and RAG at no cost, without an account, and without any data leaving the machine.
- **Options:** (a) cloud API (OpenAI, Anthropic...), (b) local models, (c) mixed: local by default, cloud for comparison.
- **Choice:** (b) 100% local. Cloud cost is **estimated** in the benchmark (measured tokens × public prices), without an account.
- **Why:** zero cost, no secrets to manage, reproducible offline. Above all, it forces us to deal with **serving** and **SLMs**, two targeted skills that a cloud API would have hidden.
- **Trade-offs:** local models (4–7B) are clearly weaker than state-of-the-art cloud models at tool calling and text-to-SQL. The qualitative results are therefore not representative of a deployment with a large model. This must be stated clearly.
- **Skill demonstrated:** Cost, model serving.

## D-02 — Python 3.12 + uv + installable package with CLI

- **Date:** 2026-10-02 · **Milestone:** J0
- **Need:** a project that installs and runs with one command, with a visible version (project convention).
- **Options:** standalone scripts + requirements.txt; Poetry; uv + pyproject.toml.
- **Choice:** uv + pyproject.toml (src/ layout), CLI entry point hello-support, single version in hello\_support/\_\_init\_\_.py (\_\_version\_\_), read dynamically by the build.
- **Why:** uv is already installed, very fast, and manages the lockfile and the venv. The src/ layout prevents importing the code without installing it. A single source of truth for the version.
- **Trade-offs:** torch with CUDA must come from the PyTorch index (cu124), not from PyPI, hence an index configuration in pyproject.toml. Added at J1, when torch became necessary.
- **Skill demonstrated:** Python, software architecture.

## D-03 — LM Studio as the local serving platform

- **Date:** 2026-10-02 · **Milestone:** J0
- **Need:** serve a local LLM behind a standard API, using the GPU, and be able to switch models without touching the code.

- **Options:** (a) LM Studio (already installed, 1ms CLI, OpenAI-compatible server), (b) Ollama (not installed), (c) vLLM (high throughput, but Linux/WSL only and heavy on 8 GB), (d) transformers directly in-process.
- **Choice:** (a) LM Studio, server at `http://localhost:1234/v1`.
- **Why:** already in place, it handles downloading (`1ms get`), GPU offload and GGUF quantization. It exposes the **OpenAI-compatible** API with *tool calling*. The application code therefore depends only on an **API contract**, not on a vendor. This is exactly the targeted "application / serving platform" separation.
- **Trade-offs:** a desktop tool, not a production server. No continuous batching and no serious throughput measurement: vLLM is noted under "going further". The server must be running (`1ms server start`) before launching the program.
- **Skill demonstrated:** Model-serving platforms.

## D-04 — Model choice

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- **Date:** 2026-10-02 · **Milestone:** J0 · **Revisable at J5** (measurements)
- **Need:** an **SLM** (3–4B) able to call tools, and a larger reference model (7B), both fitting in **8 GB of VRAM** (RTX 2070 Super).
- **SLM options:** Qwen2.5-3B-Instruct; **Qwen3-4B-Instruct-2507**; Llama-3.2-3B-Instruct; Phi-4-mini (3.8B); Gemma-3-4B (already present as `translategemma`, specialized in translation and without reliable native tool calling).
- **7B options:** `qwen2.5-coder-7b-instruct` (already present); **Qwen2.5-7B-Instruct**; Llama-3.1-8B-Instruct.
- **Choice:**
  - SLM = `qwen/qwen3-4b-2507` in **Q4\_K\_M** (~2.5 GB);
  - 7B = `qwen2.5-7b-instruct` in **Q4\_K\_M** (~4.7 GB).
- **Why:**
  - Qwen3-4B-Instruct-2507 is trained for tool calling. It is the "non-thinking" variant, so there is no reasoning block to filter out, answers are shorter and latency is lower. It is multilingual (questions in French) and was among the best public 4B models at function calling when released.
  - For the 7B, picking the **same family** (Qwen) isolates the effect of **size** in the comparison. The *Instruct* version is preferred over *Coder* for dialogue and diagnosis. It is the model named in the user's request.
  - **Q4\_K\_M** is the standard quality/size trade-off. The 7B fits entirely in VRAM (~4.7 GB + KV cache), with headroom for the embedding and reranking models (~0.5 GB). Q8 (~8 GB) would not fit.
- **Trade-offs:** 4-bit quantization slightly degrades quality, especially on SQL. Two different families would have allowed a broader but less readable comparison. The two models are not loaded at the same time: LM Studio loads them on demand (JIT), so the first call includes a loading time. It is excluded from the measurements (warm-up).
- **Implementation (J0):** `1ms get "qwen/qwen3-4b-2507@q4_k_m" -y` (LM Studio staff pick). Qwen2.5-7B-Instruct is not a staff pick: downloaded from Hugging Face with `1ms get "https://huggingface.co/lmstudio-community/Qwen2.5-7B-Instruct-GGUF@Q4_K_M" -y`.
- **Finding J0** (`hello-support smoke`, temperature 0, after warm-up): both models emit a valid `get_time({"timezone": "Europe/Brussels"})` call. SLM: hello 0.10 s, tool call 0.37 s (177 → 23 tokens). 7B: 0.10 s and 0.55 s (193 → 23 tokens). One-off measurements, not yet a benchmark (J5).
- **Skill demonstrated:** LLMs & SLMs, GPU.

## D-05 — LLM client: `openai` SDK + minimal in-house wrapper

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- **Date:** 2026-10-02 · **Milestone:** J0
- **Need:** call the model with or without tools, and retrieve content, tool calls, tokens and latency.
- **Options:** (a) `openai` SDK pointed at `localhost:1234`, (b) `langchain-openai` (ChatOpenAI), (c) raw `httpx`.
- **Choice:** (a). An `llm.py` module of a few dozen lines returns a normalized result (`content`, `tool_calls`, `prompt_tokens`, `completion_tokens`, `latency_s`, `model`).
- **Why:** we see exactly what goes over the wire and comes back, which is educational and useful for measuring. The SDK is stable and typed. LangGraph does not require LangChain for the nodes: a node is a

Python function. This limits the layers of abstraction.

- **Trade-offs:** we reimplement a bit of plumbing that ChatOpenAI + bind\_tools provide. If J3 shows this is too costly, we will switch to langchain-openai, adding an entry that supersedes this one.
- **Skill demonstrated:** Model integration, latency/cost.

## D-06 — Chunking knowledge-base sheets by section

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- **Date:** 2026-10-02 · **Milestone:** J1
- **Need:** retrieve a precise passage and **cite** it ("postgres\_connection.md, section Service status"), as the expected answer requires.
- **Options:** (a) one whole document per vector, (b) fixed-size chunking (N tokens with overlap), (c) structure-based chunking (## headings).
- **Choice:** (c). One section = one chunk, identified as doc\_id#section. The embedded text is prefixed with the sheet's title ("PostgreSQL: application cannot connect - Checks") so that the section keeps its context.
- **Why:** the sheets are short and already structured (Symptoms / Checks / Service status / Limits). Structural chunking gives stable, readable citations. Fixed-size chunking would cut in the middle of a checklist.
- **Trade-offs:** this assumes well-structured documents. On a real enterprise corpus (PDFs, wikis), hybrid chunking would be needed (structure + maximum size + overlap).
- **Skill demonstrated:** RAG, semantic retrieval.

## D-07 — Embedding model

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- **Date:** 2026-10-02 · **Milestone:** J1
- **Need:** encode questions **in French or English** and English knowledge-base sheets into the same vector space, locally, on 8 GB of VRAM shared with the LLM.
- **Options:** all-MiniLM-L6-v2 (English only); **paraphrase-multilingual-MiniLM-L12-v2** (50+ languages, 118M parameters, dim 384); multilingual-e5-base / bge-m3 (better, but 2 to 5 times larger); nomic-embed-text served by LM Studio (already present).
- **Choice:** paraphrase-multilingual-MiniLM-L12-v2 via Sentence Transformers (Hugging Face Hub, no account). Normalized vectors, cosine similarity.
- **Why:** it is the candidate proposed by the initial document. It is multilingual (the French query "connexion refusée" (connection refused) does retrieve the English sheet), lightweight (~0.5 GB of VRAM) and fast. Running it through Sentence Transformers rather than LM Studio means practicing **PyTorch / Hugging Face** directly.
- **Trade-offs:** quality is modest. Raw cosine separates poorly (0.46 to 0.52 for useful sections as well as off-topic nginx sections, see BENCH J1). This is precisely what justifies the reranker (D-09). An e5/bge model would be the natural improvement.
- **Skill demonstrated:** Embeddings, Hugging Face.

## D-08 — Vector store: embedded Chroma

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- **Date:** 2026-10-02 · **Milestone:** J1
- **Need:** store and query vectors (top-k by similarity) without a server to administer.
- **Options:** (a) in-memory NumPy (initial proposal: brute-force cosine), (b) embedded **Chroma** (PersistentClient, HNSW index), (c) sqlite-vec (SQLite extension), (d) PostgreSQL + pgvector (WSL/Docker), (e) FAISS.
- **Choice:** (b) Chroma 1.5, kb\_sections collection in **cosine** space, persisted in .chroma/ (gitignored). Embeddings are computed by our code and passed to Chroma, which does not choose the model. The index is rebuilt only if the **fingerprint** (hash of the model + section contents) changes.
- **Why:** it is a real vector database (ANN index, metadata, persistence), installed with pip, without Docker. It covers the "vector database" skill at low cost. Supplying our own vectors keeps control of the model (D-07) and allows changing it without depending on Chroma's built-in embedding functions. With the fingerprint, we avoid recomputing at every startup while guaranteeing the index is never stale.

- **Trade-offs:** with 12 vectors, an ANN index brings **no** gain: NumPy brute force would be just as fast. The choice is educational, and this must be said. Chroma adds many dependencies. pgvector would be closer to an "AI + database" context: it is placed under "going further". Planned fallback: NumPy cosine if Chroma caused problems (it did not).
- **Skill demonstrated:** Vector search, vector databases.

## D-09 — Cross-encoder reranking

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- **Date:** 2026-10-02 · **Milestone:** J1
- **Need:** improve the order of the passages returned to the agent (only 2, so the ranking matters), and detect that **no** passage matches.
- **Options:** (a) no reranking, (b) Sentence Transformers **cross-encoder**, (c) LLM reranker (ask the model to rank), (d) bge-reranker-v2-m3 (more accurate, ~570M parameters).
- **Choice:** (b) cross-encoder/mmarco-mMiniLMv2-L12-H384-v1 (multilingual, lightweight). Vector search top-5, rerank of the 5 (question, passage) pairs, top-2 kept. Each hit keeps its original vector rank to show the effect of reranking. Threshold `RELEVANCE_THRESHOLD = -5` on the cross-encoder score: below it, the passage is marked "off-topic".
- **Why:** this is the classic *retrieve & rerank* pattern. The bi-encoder is fast but coarse; the cross-encoder reads the question and the passage together and ranks better. Measured (BENCH J1): on "connexion refusée postgres", the *Symptoms* section moves from 5th to 2nd place, ahead of two nginx sections. On a question outside the knowledge base (Kafka), all scores drop to  $\leq -9.2$  while useful passages are  $\geq -4.3$ . This signal is far more discriminating than cosine (0.12–0.21). An LLM reranker would have cost one model call per query.
- **Trade-offs:** about 0.1–0.3 s more per query. The threshold is **empirical** and calibrated on 4 queries: it is a coarse filter, not a guarantee of precision. It should be calibrated on a question set (see "going further": RAG evaluation).
- **Revision (J3):** `search_docs` returns the **top-3** (instead of top-2). With 2 passages, the reranker put "Limits" first and the "Service status" section disappeared from the context. Only the diagnostic command `hello-support search` keeps a default of `--top-n 2`.
- **Skill demonstrated:** Reranking.

## D-10 — GPU inference and warm-up

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- **Date:** 2026-10-02 · **Milestone:** J1
- **Need:** use the available GPU and measure a representative query latency.
- **Options:** CPU only; GPU via torch CUDA; embeddings served by LM Studio.
- **Choice:** torch **2.6.0+cu124** pulled from the PyTorch index (configured in `pyproject.toml`, since PyPI only provides the CPU build on Windows). Device chosen automatically (`cuda`, otherwise `cpu`) and displayed. A **warm-up** query is run when the Retriever loads.
- **Why:** the GPU is there (RTX 2070 Super). Both small models fit on it (~1 GB) alongside the LLM served by LM Studio. Measured: the first query costs ~2.9 s (CUDA kernel initialization) versus ~0.5 s afterwards. Warm-up removes this one-time cost from the per-question latency.
- **Trade-offs:** heavy torch CUDA download (~2.5 GB), slow `sentence_transformers` import (~15 s cold on Windows), total load ~27 s. Acceptable because it is paid once per tool-server session (J2). On CPU, the project works too, just more slowly.
- **Skill demonstrated:** GPU, PyTorch, latency.

## D-11 — MCP: separate tool server, stdio transport, Python SDK v2

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- **Date:** 2026-10-02 · **Milestone:** J2
- **Need:** expose the tools to the agents through a **standard protocol**, so they can be tested on their own, reused by another client (Inspector, IDE, another agent), and moved without touching the agents' code.
- **Options:** (a) Python functions called directly by the agents, (b) in-house REST API, (c) **MCP** over stdio (local subprocess), (d) MCP over *streamable HTTP* (network server).

- **Choice:** (c) with the official mcp SDK **2.2** (MCPServer, the new name of FastMCP in v2). Entry point `hello-support-mcp`. The host will launch the server as a subprocess (J3). Tests use the same server **in memory** via `mcp.Client(server)`.
- **Why:** MCP is one of the targeted skills (integrating enterprise tools for agents). stdio needs no port and no auth and is sufficient locally. The input JSON schema is generated from Python annotations. Verified with **MCP Inspector** (`npx @modelcontextprotocol/inspector --cli hello-support-mcp --method tools/list` then `tools/call` for each of the 3 tools).
- **Trade-offs:** one subprocess per session and one more protocol to debug. SDK v2 is recent: online examples (FastMCP, camelCase inputSchema) no longer apply as is. The Inspector in CLI mode handles `-m module` arguments poorly, hence the dedicated entry point. The HTTP transport (a first "distributed" step) is left to "going further".
- **Skill demonstrated:** MCP, tool integration.

## D-12 — Tool contracts: strict types, explicit errors

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- **Date:** 2026-10-02 · **Milestone:** J2
- **Need:** the model **proposes** calls that can be wrong (unknown service, invalid SQL, empty query). These errors must be rejected cleanly and remain usable by the model.
- **Options:** validation in each agent; validation on the tool-server side; both.
- **Choice:** **server-side** validation, through types: `service_name: Literal["postgres", "nginx", "redis"]` produces an enum in the MCP schema, which the model sees and Pydantic checks. Business errors raise `ToolError`, returned to the client as `is_error=true` with a clear message (e.g. "SQL rejected: only SELECT queries are allowed"). The host only checks that the arguments are valid JSON; types and enums are validated by the MCP server. For an unknown service, the model therefore receives the Pydantic literal error ("Input should be 'postgres', 'nginx' or 'redis'"): the "unknown service" message of `data_store.service_status` is a safety net for direct Python callers and is not reachable through MCP. Tool descriptions (docstrings) explain the output format and usage, for example `relevant=false`.
- **Why:** a single place is authoritative, whatever the client. The error returned to the model lets it **correct itself once** (J3) instead of crashing. The enum schema reduces errors at the source, which matters for an SLM.
- **Trade-offs:** Pydantic validation messages are verbose for a small model. The server does not know about **call limits**: that is the orchestrator's job (J3).
- **Skill demonstrated:** Tool calling, reliability.

## D-13 — Read-only SQL tool: defense in depth

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- **Date:** 2026-10-02 · **Milestone:** J2
- **Need:** let the agent **write SQL** (text-to-SQL, like an "enterprise data agent") without it being able to modify or exfiltrate anything, or lock the database.
- **Options:** (a) parameterized tool (`get_incidents(service, days)`), no free-form SQL; (b) free-form SQL with a regex filter; (c) free-form SQL with **several independent layers**.
- **Choice:** (c), in `sql_guard.py`:
  1. static check: a single statement, starting with `SELECT/WITH`, with no write or admin keyword (`DELETE`, `PRAGMA`, `ATTACH`, `load_extension...`);
  2. wrapped query: `SELECT * FROM (<sql>) LIMIT 51`, then truncated to 50 rows with a truncated flag;
  3. SQLite connection in `mode=ro`: the engine refuses any write;
  4. SQLite **authorizer**: only `SELECT/READ/FUNCTION/RECURSIVE` actions are allowed;
  5. **progress handler**: the query is aborted after 2 s.

The table schema and a date-filter example are in the tool description. - **Why:** (a) would be safer, but does not practice text-to-SQL, which is the core of a data agent. A regex alone can be bypassed: each layer covers the blind spots of the others, and layers 3–4 are guaranteed by the engine, not by our code. SQL errors (unknown column...) are returned as is so the model can fix its query. Tested: `DELETE`, `SELECT 1`; `DROP...`, `PRAGMA`, `ATTACH`, `load_extension` are

rejected; the row limit is enforced (tests/test\_sql\_guard.py). - **Trade-offs**: the regex can reject a legitimate query containing one of these words in a string (e.g. summary LIKE '%update%'). For the same reason, a ; inside a string literal (e.g. summary LIKE '%a;b%') is rejected as a second statement (tests/test\_guardrails.py::test\_semicolon\_inside\_a\_string\_literal\_is\_rejected). Acceptable here, to be fixed with a real SQL parser (sqlglot) if needed. Nothing prevents a **wrong but valid** query: its correctness is evaluated at J4 (case C4). On a real enterprise database, a dedicated read-only SQL role and restricted views would also be needed. - **Skill demonstrated**: Enterprise data agents, SQL engines.

## D-14 — Data: generated SQLite incident database, simulated JSON scenarios

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- **Date**: 2026-10-02 · **Milestone**: J2
- **Need**: realistic structured data for analysis questions ("combien d'incidents postgres ces 30 derniers jours ?" (how many postgres incidents in the last 30 days?)) and **simulated**, controllable service states for cases C1/C2/C6.
- **Data options**: real PostgreSQL (WSL/Docker), DuckDB, **SQLite** (stdlib). **Status options**: query real services; a JSON scenarios file.
- **Choice**: incidents table (14 rows: service, date, severity, summary, resolved) generated by seed\_incidents() with dates **relative to the seed day**. The data/incidents.db file is gitignored. Statuses come from scenarios.json (stopped, running, redis\_down, tool\_error). The scenario is chosen in this order: the content of the file named by HS\_SCENARIO\_FILE (used by the web demo, D-31), then --scenario / HS\_SCENARIO, then the default stopped. Each response carries simulated: true and the scenario name.
- **Revision (v1.11.2)**: the database was first seeded **once**, so its dates stayed relative to the day of the first run. About 12 days later, "the last 30 days" would have dropped from 3 to 2 postgres incidents and the C4 check would have failed for a correct answer (review F-1). The seed day is now stored in PRAGMA user\_version (YYYYMMDD, invisible to the model's queries), and ensure\_incidents\_db re-seeds the database when that day differs from today or is missing (file created before v1.11.2). If another process holds the file open, the existing data is kept for that query. Tests:  
tests/test\_guardrails.py::test\_database\_is\_reseeded\_when\_the\_day\_changes and  
::test\_database\_without\_a\_seed\_day\_is\_reseeded.
- **Why**: SQLite is in the stdlib, serverless, and a real SQL engine (dates, aggregates, recursive CTEs, authorizer). With dates relative to a seed day that is refreshed daily, "the last 30 days" always has the same expected answer (3 postgres incidents), which makes case C4 verifiable. Scenarios make behavior **deterministic** and allow simulating a tool failure (C6). Since nothing real is touched, no dangerous action is possible.
- **Trade-offs**: SQLite is not an enterprise engine: no roles, no write concurrency, no distributed plan. 14 rows do not test performance. PostgreSQL + pgvector is under "going further".
- **Skill demonstrated**: Database integration, data analysis.

## D-15 — Loading the RAG in the background in the MCP server

---

- **Date**: 2026-10-02 · **Milestone**: J2
- **Need**: the Retriever takes ~30 s to load. If it is loaded before starting the server, the MCP handshake exceeds the client timeout (15 s by default for the Inspector).
- **Options**: load at startup (blocking); lazy load on first call; **load in a thread at startup**, with the first search\_docs waiting for it to finish.
- **Choice**: the thread. The handshake is immediate, the get\_service\_status and query\_incidents tools are available right away, and search\_docs waits at most 180 s. Loading errors are returned as ToolError.
- **Why**: in general the documentalist calls search\_docs first, after one LLM call: loading happens during that time instead of adding to latency.
- **Trade-offs**: a bit of concurrency to manage (an Event). Measured via the Inspector: a cold search\_docs call takes ~54 s end to end (npx startup + process + loading). This is still paid once per session.
- **Skill demonstrated**: Latency.

## D-16 — Orchestration: LangGraph StateGraph, nodes = Python functions

---

- **Date:** 2026-10-02 · **Milestone:** J3
- **Need:** chain steps (triage → search → diagnosis), with conditional branches, shared state and guaranteed termination.
- **Options:** (a) a hand-written Python loop, (b) **LangGraph** (state graph), (c) a single ReAct agent (`create_react_agent`) that decides everything, (d) CrewAI / AutoGen (conversational multi-agent).
- **Choice:** (b) LangGraph 1.2, graph `START → triage → documentalist → technician → END`, with a **conditional edge**: history questions (`history`) go straight to the technician, with no document search. Another one exits to `END` if the documentalist fails. Nodes are plain `async` functions: `LangChain` is not needed (D-05). `recursion_limit=10` serves as a safety net.
- **Why:** it is the target agent framework. It clearly separates the **code-defined workflow** (structure, branches, tool policy) from the **model's decisions** (tool arguments, SQL, writing), which the initial document already recommended. A single ReAct agent leaves everything to the model, which proved unreliable with 4–7B models (D-17).
- **Trade-offs:** for 3 linear nodes, LangGraph is almost oversized. Its value shows up with branches, checkpoints and human-in-the-loop ("going further"). Loading LangGraph adds dependencies.
- **Skill demonstrated:** Orchestration, LangGraph.
- **Revision (v1.10.0, D-30):** the conditional edge also sends `out_of_scope` and `vague` straight to the technician; only `malfunction` and `documentation` go through the documentalist.

## D-17 — Triage via structured output + tool policy enforced by code

---

- **Date:** 2026-10-02 · **Milestone:** J3 · **Decision driven by a measured failure**
- **Need:** the technician must **observe** (call `get_service_status`) when an outage is reported, and must not do so for a documentation question.
- **Finding:** the first version left this choice to the model (`tool_choice="auto"`, rules in the prompt). Measured on the technician's actual input: **0/3** calls for Qwen2.5-7B **and** for Qwen3-4B, whatever the layout (reminder at the end of the message, no brief, question last). Both models answered conditionally ("si le service est arrêté..." — "if the service is stopped...") since the sheet describes both cases. The 7B went as far as telling the user to call `get_service_status` themselves.
- **Options:** (a) more prompt engineering; (b) systematic tool call; (c) **router**: a short LLM call that classifies the intent into constrained JSON, followed by a tool policy decided by the code; (d) a bigger model.
- **Choice:** (c). triage node: strict JSON Schema `response_format → {intent: malfunction|history|documentation|out_of_scope|vague, service}`. The code applies `TECHNICIAN_POLICY`:
  - `malfunction` → Only `get_service_status` is offered, and one call is **required**;
  - `history` → `query_incidents` is **required**, the model writes the SQL;
  - `documentation` → no tool;
  - `out_of_scope / vague` → no tool and an explicit instruction (state the limit / ask a question).

An outage with no known service is reclassified as `vague`.

**What "required" means exactly:** the call with tools is forced (`tool_choice="required"`) and, if the model still answers in text, that answer is discarded and the model is retried once (D-18). If it answers in text a second time, the answer is **accepted** and the status stays `done`; since v1.11.2 the code appends the error "technician: a tool call was required but never made; answer accepted without observation" and emits a `limit` event with the same text, so the missing observation is visible in the trace and in the answer's errors (`tests/test_guardrails.py::test_required_tool_never_called_is_traced`). The code therefore guarantees the attempt and its traceability, not the observation itself.

**Triage failure:** if the triage call fails or returns invalid JSON, the route falls back to `documentation` with no service, and an `error` event is emitted. When LM Studio is down, the documentalist then fails too: the answer is the fixed `FALLBACK_ANSWER`, the status is `failed`, and `hello-support ask` exits with code 1 (0 when the status is `done`). - **Why:**

measured on 6 typical questions, triage costs  $\sim 0.3$  s (after loading) and correctly classifies the key cases. Qwen2.5-7B: 4/6, with both "errors" (Kafka  $\rightarrow$  vague, "ça marche pas" ("it doesn't work")  $\rightarrow$  outage with no service) still leading to the right behavior. Qwen3-4B: 4/6, but it confused "outage + what should I check?" with documentation, fixed by a clarification in the prompt. These 4/6 figures (and the 0/3 of the finding above) were measured ad hoc during development; the raw output is not archived in the repository. The model still decides, but **on a narrow, constrained task**, and the code enforces the tool policy that follows from it (with the fall-through above). This is a common *router + policy* pattern in production. - **Trade-offs**: one more LLM call per request. The classification can be wrong, and the error propagates (no going back). The 5 intents are hard-coded. - **Skill demonstrated**: AI agents, intelligent workflows.

## D-18 — Bounded agent loop and measured guardrails

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- **Date**: 2026-10-02 · **Milestone**: J3
- **Need**: no infinite loop, a bounded cost, tool errors that do not crash the program, and no invented observation.
- **Choice** (`agents.run_agent`, shared by both agents):
  - at most 3 LLM calls per agent. The **last call is made without tools**, which forces a text answer;
  - per-tool budgets: `search_docs`  $\leq 2$ , `get_service_status`  $\leq 1$ , `query_incidents`  $\leq 2$ . Exceeding one returns an error message to the model, not an exception;
  - **deduplication** of identical calls and **at most 3 calls per step**. Measured reason: the 7B, forced to call a failing tool (`tool_error` scenario), emitted **46 identical calls** in 1,024 tokens (18 s);
  - `max_tokens` = 300 for a step with tools, 700 for a final answer;
  - `tool_choice="required"` **checked by the code**: LM Studio does not strictly enforce it. Once, the 7B answered in text and **invented** a "en cours d'exécution" ("running") status without calling the tool. An answer without the required observation is therefore rejected and the model retried once. A second text answer is accepted but flagged as unobserved (error and limit event, D-17);
  - invalid JSON arguments and unauthorized tools: refused with a message;
  - HTTP timeout for LLM calls (120 s) and tool calls (240 s).
- **Why**: the model is not trusted to stop or to respect an API constraint. Each guardrail above comes from a behavior **observed** during J3.
- **Trade-offs**: the limits are scope choices, not optima. A rejected answer costs one LLM call. Reasoning quality is not guaranteed: in case C4, the 7B computed `MAX(resolved)` and still concluded "le dernier n'est pas résolu" ("the last one is not resolved"). This is correct by chance, but not deduced from the result (J4 evaluation).
- **Skill demonstrated**: Reliability, workflow execution.

## D-19 — Explicit shared state and JSON trace

---

- **Date**: 2026-10-02 · **Milestone**: J3
- **Need**: know at any time what each step saw and decided, replay and measure a run.
- **Choice**: State (TypedDict): `question` (never modified), `model`, `scenario`, `route`, `evidence` (relevant passages only, deduplicated), `brief`, `observations`, `counters`, `status`, `errors` and `trace` (the last two with an `operator.add` reducer: nodes **append**, they do not overwrite), `answer`. Each event (start when an agent's step begins, `llm`, `tool_call`, `tool_result`, `limit`, `error`; the triage `llm` event also carries the route) is timestamped, displayed live and written to `runs/<timestamp>.json` with the metrics (total duration, LLM calls and time, tokens in/out, tool time). The `runs/` folder is gitignored.
- **Why**: updates go explicitly through node return values, which is LangGraph's model. Displaying the **observable decisions** (tool, arguments, result) rather than an internal "reasoning" follows the initial document's recommendation. The trace was used to diagnose each of the problems in D-17/D-18.
- **Trade-offs**: the state lives in memory for the duration of a request. There is no recovery after a crash (LangGraph checkpointing = "going further"). The trace contains the full texts, which could be a problem with sensitive data in a real-world context.
- **Skill demonstrated**: State management, latency/cost.

## D-20 — Host-side MCP client: the `mcp` SDK directly rather than `langchain-mcp-adapters`

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- **Date:** 2026-10-02 · **Milestone:** J3
- **Need:** start the tool server, list its tools in the format the LLM expects, and call them.
- **Options:** `langchain-mcp-adapters` (converts to LangChain tools); the SDK's `mcp.Client`.
- **Choice:** `ToolBox` ( $\approx 80$  lines): `mcp.Client(StdioServerParameters(...))` starts python `-m hello_support.mcp_server` with `HS_SCENARIO`, converts `list_tools()` to the OpenAI `tools` format (the MCP JSON schema is reused as is) and decodes the results. A list is returned as `structured_content = {"result": [...]}` or as one text block per item. The same `ToolBox` accepts an **in-memory** server for tests.
- **Why:** without LangChain in the loop (D-05), the adapter adds nothing. The direct SDK shows what an MCP host actually does: *spawn, handshake, list, call*.
- **Trade-offs:** result decoding has to be handled by hand. A first bug (list returned in several blocks) was found on the first real run.
- **Skill demonstrated:** MCP, tool integration.

## D-21 — Validation: 6 executable cases with automatic checks

---

- **Date:** 2026-10-02 · **Milestone:** J4
- **Need:** know whether the system "works" on the spec's cases, re-check it after every prompt or model change, and compare two models on the same basis.
- **Options:** (a) read the answers by hand; (b) **LLM-as-a-judge** (a model grades the answers); (c) **deterministic checks** on the final state (route, tools called, results, keywords and citations in the answer); (d) an evaluation framework (Ragas, DeepEval...).
- **Choice:** (c), in `cases.py`: 6 cases, each with 5 to 7 named checks (e.g. "observed status = stopped", "no status call", "last incident read (ORDER BY...)", "exact citations", "in French"). The `hello-support bench` command: `N runs × models`, a Markdown report (`docs/bench/`) with **all the answers**, and raw JSON (`runs/`). Unit tests (pytest) check the code with a **scripted** LLM; the benchmark checks the system with the **real** models.
- **Why:** it is reproducible, free and explainable. A failing check points to a specific defect. The benchmark immediately revealed two defects that were invisible in demos (D-22, D-23). A local LLM judge (4–7B) would be less reliable than the answers it grades, and a framework would add dependencies for 6 cases.
- **Trade-offs:** keyword checks are **coarse**. A good answer can fail (unexpected wording), a bad one can pass. They do not measure the quality of the advice. Hence full answers are kept for human review. 6 cases do not make a statistical evaluation.
- **Skill demonstrated:** Prototype → production, evaluation.

## D-22 — Deterministic post-processing: normalized citations + simulation notice

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- **Date:** 2026-10-02 · **Milestone:** J4 · **Decision driven by a measured failure**
- **Finding** (J4 benchmark, 7B): despite the "copy the labels exactly" instruction, the model writes `postgres_connection.md#Service_status` (underscore). It also omits the "simulated" notice required by C1.
- **Options:** strengthen the prompt even more; change the identifier format; **post-process the answer in code**.
- **Choice:** `postprocess.py`:
  - `normalize_citations` maps each `doc.md#Section` citation back to the knowledge base's canonical identifier (case- and \_-insensitive) and flags unknown citations in the trace, without removing them;
  - `simulation_footer` appends at the end of the answer a line built **from the tool results**: "Statut observé en simulation (scénario « stopped ») : postgres = stopped. Aucune action corrective n'a été exécutée." ("Status observed in simulation ("stopped" scenario): postgres = stopped. No corrective action was executed.") The sentence is in French if the question is.
- **Why:** it is the same principle as D-17. What the code **knows** (the status comes from a simulation, nothing was executed, the list of existing sections), the code says, instead of hoping the model will say it. This makes the answer verifiable and consistent with the expected answer in the initial document.

- **Revision (v1.12.0):** the same principle now covers the **answer language**. Testing the README quick start showed the 7B answering an English question in French every time, although the prompt said "same language as the question" (with a French example). The code now detects the language (same heuristic as the footer) and ends the technician input with "Answer language: English|French". Test: `tests/test_workflow_fake_llm.py::test_the_code_tells_the_technician_the_answer_language`; measured in [BENCH](#), section v1.12.0.
- **Trade-offs:** C1's "says simulated" check is now satisfied **by construction**. It tests the system, not the model, and that must be stated. Language detection is a heuristic (French function words).
- **Skill demonstrated:** Reliability.

## D-23 — Data questions: all queries in the same step

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- **Date:** 2026-10-02 · **Milestone:** J4 · **Decision driven by a measured failure**
- **Finding:** in case C4 ("combien... et le dernier est-il résolu ?" — "how many... and is the last one resolved?"), the 7B ran a `COUNT(*)`, then **wrote** "Now let's check the status of the last incident" and **invented** the answer instead of calling the tool a 2nd time. An earlier version drew its conclusion from a `MAX(resolved)` that cannot answer the question.
- **Choice:** the history instruction asks for **all queries in the same step** (parallel tool calls), one query per part of the question, with the example `ORDER BY started_at DESC LIMIT 1`. It also forbids describing a query that was not run. The C4 check verifies that a query reading the last incident was actually **executed**.
- **Why:** the model hallucinates when it "narrates" the next step instead of doing it. Having it plan the queries all at once removes that intermediate step. Measured result: 2 queries issued in parallel, conclusion "3 incidents, the last one unresolved" **deduced from the returned rows**.
- **Trade-offs:** this relies on the model's ability to make parallel calls. A question requiring a query that depends on the result of another (true multi-step analysis) would not be covered: it would then need an explicit plan or a larger step budget.
- **Skill demonstrated:** Enterprise data agents.

## D-24 — Triage: explicit rules + examples, measured on held-out paraphrases

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- **Date:** 2026-10-02 · **Milestone:** J4 · **Decision driven by a measured failure**
- **Finding** (J4 benchmark #1): the SLM classified "Mon application ne parvient plus à se connecter à PostgreSQL. Que dois-je vérifier ?" ("My application can no longer connect to PostgreSQL. What should I check?") as documentation **3 times out of 3**. It therefore never observed the status, hence  $C1/C2 = 0/3$ . Adding examples alone (*few-shot*) fixed the SLM on the paraphrases, but **degraded the 7B**: "Mon Kafka est lent" ("My Kafka is slow") → `malfunction/nginx`, "My MongoDB replica set is down" → `malfunction/redis`. This is dangerous, since it would observe a service unrelated to the question.
- **Choice:** triage prompt = definitions + **2 explicit rules** (another product mentioned → `out_of_scope`, never mapped to a known service; an outage happening now → `malfunction`, even if the sentence ends with "what should I check?") + **6 examples** worded differently from the benchmark questions.
- **Measurements:** 12 questions = the 6 benchmark ones + **6 held-out paraphrases** (never seen during tuning): **12/12 for both models**, versus 11/12 (SLM) and 10/12 (7B) with examples alone. These figures were measured ad hoc; the question set and the raw output are not archived in the repository, so they cannot be replayed with a shipped command.
- **Why:** an example without a rule can be over-generalized (the 7B learned "unknown product ⇒ pick a known service"). The rule sets the boundary, the examples show the format. Testing on unseen paraphrases avoids tuning the prompt for the benchmark questions only (overfitting).
- **Trade-offs:** 12 questions is still a small sample. Every prompt change must be re-measured, since a gain on one model can be a loss on the other (as observed here).
- **Skill demonstrated:** Evaluate AI tech, SLMs.

## D-25 — ADR: default model = Qwen2.5-7B-Instruct; the SLM remains a measured option

- **Date:** 2026-10-02 · **Milestone:** J5 · **Status:** Accepted. It confirms D-04 with measurements.
- **Context:** two models served locally by LM Studio, same prompts, same code, 6 cases × 3 runs (BENCH J4), plus a serving throughput measurement (BENCH J5).
- **Options:** (a) SLM Qwen3-4B-Instruct-2507 by default; (b) **7B by default**; (c) hybrid: SLM for triage (narrow task, constrained output), 7B for the answer.
- **Data:**

| Criterion                                                   | SLM 4B                                                 | 7B                                                                                                                                                                                                                |
|-------------------------------------------------------------|--------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Cases passed (6 × 3)                                        | 15/18                                                  | <b>18/18</b>                                                                                                                                                                                                      |
| Serious defect observed                                     | invents source identifiers when a tool fails (C6, 3/3) | sometimes presents the sheet's symptoms as "observed logs" (not detected by the checks); asserted an unobserved Redis status once when the status tool failed (C6, 1 of 10 runs in the v1.10.0 control, see D-30) |
| Triage (12 questions incl. 6 held out)                      | 12/12                                                  | 12/12                                                                                                                                                                                                             |
| p50 latency per request (warm)                              | <b>6.0 s</b>                                           | 9.0 s                                                                                                                                                                                                             |
| Generation throughput, 1 request at a time                  | <b>107 tokens/s</b>                                    | 71 tokens/s                                                                                                                                                                                                       |
| Aggregate throughput, 4 concurrent requests                 | <b>173 tokens/s</b>                                    | 128 tokens/s                                                                                                                                                                                                      |
| VRAM (Q4_K_M)                                               | ~2.5 GB                                                | ~4.7 GB                                                                                                                                                                                                           |
| Estimated cost if API (gpt-4o-mini / Haiku 4.5), /1000 req. | ~\$0.62 / ~\$4.5                                       | ~\$0.62 / ~\$4.6                                                                                                                                                                                                  |

The table above is the J4/J5 measurement (v1.7.0 code). **Update (v1.11.2, 2026-10-04):** full benchmark 6 cases × 3 with the pinned LM Studio settings:

| Criterion                            | SLM 4B                                                 | 7B                                                                   |
|--------------------------------------|--------------------------------------------------------|----------------------------------------------------------------------|
| Cases passed (6 × 3)                 | 16/18                                                  | 16/18 raw, <b>18/18</b> re-scored with the fixed status check (D-30) |
| Checks passed                        | 105/108                                                | 106/108 raw, 108/108 re-scored                                       |
| C6 (tool failure)                    | 1/3 (2× "says the check failed", 1× "exact citations") | 3/3 re-scored; control 10/10                                         |
| p50 / max latency per request (warm) | <b>4.4 s</b> / 7.23 s                                  | 8.47 s / 15.8 s                                                      |
| LLM calls per request                | 4.28                                                   | 4.0                                                                  |
| Tokens in / out per request          | 2,658 / 351                                            | 2,462 / 460                                                          |

Raw reports: docs/bench/bench-20261004-185627.md (4B), bench-20261004-190049.md (7B) and bench-20261004-190503.md (7B C6 control, 10 runs). - **Decision:** (b) 7B by default (HS\_MODEL\_DEFAULT=large). The SLM remains selectable (--model s1m) and is the best candidate for triage if latency becomes the priority. - **Why:** for a support assistant, an **invented source** is more serious than an answer that is 3 s slower. The SLM is faster on every metric (~1.5×) and just as good at triage. It only loses on robustness when a tool fails, but that is the case that matters in production. The **local** cost is zero in both cases, and the estimated cloud cost is identical, because tokens are mostly input (RAG context) and the volumes are similar. Model size therefore does not change the bill: context length is what matters. - **Consequences / next steps:** (c) is the next logical experiment: triage by the SLM, answer by the 7B. It would need to be measured with the same benchmark before adoption. With a larger (cloud) model, most guardrails would remain useful: they protect against classes of errors, not against a specific model. - **Skill demonstrated:** LLMs & SLMs, latency/cost, evaluate → plan.

## D-26 — Measurement method: latency, throughput, cost

- **Date:** 2026-10-02 · **Milestone:** J5
- **Need:** discuss latency, throughput and cost with **measured figures** and a method that can be explained.
- **Choice:**
  - **latency:** per request and warm (MCP session and model already loaded), p50/max over 18 requests per model. **Cold start** (20–26 s: torch, embedding models, Chroma, LM Studio loading) is measured separately, because it is paid once per session;
  - **breakdown** via the trace: LLM time, tool time, number of calls;
  - **throughput:** (1) generation throughput in the benchmark (tokens produced / LLM time, prompt included); (2) hello-support throughput: the same request repeated 8 times at concurrency 1 then 4, to measure aggregate throughput (tokens/s, requests/min) and individual latency;
  - **cost:** local = €0 (electricity not counted). Cloud = **estimate:** measured average tokens in/out × indicative public prices, with no account and no call, and prices to be re-checked before quoting them.
- **Key results:**
  - ~2,600–2,800 input tokens for ~350–390 output tokens per request: the **context** (prompts, RAG passages, tool results) dominates the cost, so that is where to optimize;
  - concurrency 4 multiplies aggregate throughput by 1.6–1.8, but doubles individual latency: this is the classic throughput / latency trade-off of serving;
  - loading the RAG (~14–27 s) is the largest item in cold latency.
- **Trade-offs:** a single machine, few runs, temperature 0: no real statistical distribution. Throughput is measured on LM Studio, not on a production server (vLLM, continuous batching: "going further"). Prices change.
- **Skill demonstrated:** Latency, throughput, cost.

## D-27 — Code structure

- **Date:** 2026-10-02 · **Milestone:** J5 (review)
- **Choice:** a src/hello\_support/ package of single-responsibility modules:

| Module                       | Role                                                                      | Depends on                             |
|------------------------------|---------------------------------------------------------------------------|----------------------------------------|
| cli.py                       | smoke, search, ask, bench, throughput, web commands, display              | everything else (lazy imports)         |
| webapp.py                    | web demo: Starlette app, Server-Sent Events, single-question queue (D-31) | workflow, toolbox, starlette           |
| static/                      | the demo page (HTML, CSS, JS, no framework)                               | —                                      |
| workflow.py                  | LangGraph graph, state, trace export                                      | agents, toolbox, postprocess           |
| agents.py                    | triage, prompts, tool policy, bounded loop                                | llm, toolbox                           |
| llm.py                       | OpenAI-compatible client, measurements                                    | openai                                 |
| toolbox.py                   | MCP host (spawn, list, call)                                              | mcp                                    |
| mcp_server.py                | MCP server, 3 tools                                                       | retrieval, data_store, sql_guard       |
| retrieval.py                 | chunking, embeddings, Chroma, reranking                                   | torch, sentence-transformers, chromadb |
| sql_guard.py / data_store.py | read-only SQL; simulated data and SQLite                                  | stdlib                                 |
| postprocess.py               | citations, simulation notice                                              | retrieval (list of sections)           |
| cases.py / benchmark.py      | validation cases, measurements, reports                                   | workflow                               |
| config.py                    | environment variables                                                     | python-dotenv                          |

- **Why:** the **MCP** boundary physically separates the agents (host process) from the tools and data (subprocess): the agents do not know how a tool is built. Heavy dependencies (torch, Chroma) are loaded only by the tool server and the search command. Tests and the CLI stay fast. Everything (code, prompts, comments and documentation) is in English (UR-006); only the validation questions are French test data.
- **Trade-offs:** about 1,950 lines of Python and 420 of HTML/CSS/JS in `src/` (+ about 850 lines of tests) for a "hello world", a good part of which is guardrails and measurement. Prompts are Python constants, not separately versioned files.
- **Skill demonstrated:** Software architecture, Python.

## D-28 — Testing strategy

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- **Date:** 2026-10-02 · **Milestone:** J5 (review)
- **Choice:** three levels.
  1. **Unit, offline** (`uv run pytest`, 108 tests in ~16 s at v1.11.2): sheet chunking, SQL guardrails (rejection of DELETE / DROP / PRAGMA / ATTACH / load\_extension, row limit), MCP contracts via a **real MCP client connected in memory**, agent loop and full graph with a **scripted LLM** (budgets, deduplication, last call without tools, retry when a required tool is ignored, tool error), post-processing, case checks; the web demo (`tests/test_web.py`); the user requirements, including the language and diagram checks of the documentation (`tests/test_user_requirements.py`); and `tests/test_guardrails.py`, added after the review for the guardrails that had no test: per-step call cap, invalid JSON arguments, tool not offered, max\_tokens caps, tool timeout, SQLite authorizer, SQL timeout (plus the v1.11.2 fixes: required tool never called, ; in a literal, daily re-seed, web path).
  2. **Integration with models** (`pytest -m models`): end-to-end RAG on GPU.
  3. **System with the real LLMs** (`hello-support bench`): 6 cases × N × models, automatic checks and archived answers.
- **Why:** what is deterministic (guardrails, contracts, orchestration) is tested deterministically. What depends on the model is **measured**, not "tested green". The scripted LLM makes it possible to reproduce exactly the faulty behaviors observed (answer without the required tool, 46 duplicated calls) and to lock in the fixes.
- **Trade-offs:** no CI (out of scope). The benchmark is not run automatically: it needs LM Studio and ~6 min. Keyword checks remain coarse (D-21).
- **Skill demonstrated:** Production-grade, reliability.

## D-29 — Agents, router and code: who decides what, and why there is no loop between agents

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- **Date:** 2026-10-04 · **Milestone:** documentation (no behavior change)
- **Need:** make the nature of each block readable. A reader of the diagram asked: "if there are agents, where are the loops?", "is triage an agent?"
- **Finding (code):**
  - **Agents** = Documentalist and Technician. `run_agent` loops the LLM over the tools: the model chooses the tool and its arguments, the code executes it and returns the result. The loop makes  $\leq 3$  LLM calls (`MAX_LLM_CALLS`), the last one without tools,  $\leq 3$  tool calls per step. A required tool is retried once if the model ignored it (a second text answer is accepted but flagged, D-17), and the loop stops as soon as there is a text answer.
  - **Triage = router:** 1 LLM call with a JSON Schema `response_format` (intent, service), with no tool and no loop. The code derives the graph branch and the tool policy from it (D-17).
  - **Deterministic code:** loop controls (D-18), post-processing (D-22), reclassification of an outage with no known service as vague.
- **Options:** (a) a multi-agent graph with back-edges (the technician can send back to the documentalist, or a supervisor re-arbitrates); (b) **a linear graph, with loops kept inside the agents.**
- **Choice:** (b). Each LangGraph node runs at most once per question.
- **Why:** the need is sequential: find the sources, then diagnose. A back-edge between agents would add LLM calls (latency, tokens), a risk of ping-pong, and behavior that is harder to test and bound, with no measured

- gain on the 6 cases. The internal loops are enough to fix a failed query or tool call. The limits (3 LLM calls, tool budgets, recursion\_limit=10) guarantee termination.
- **Trade-offs:** if the passages found are poor, the technician cannot request a new search. It flags this in its answer. A documentalist ↔ technician loop, or a supervisor, would be the natural evolution, to be measured with the same benchmark.
  - **Representation:** diagrams with a legend (agent ■, router ■, code ■, MCP tool ■, model ■) and a zoom on an agent's loop (README, summary above, SPEC §4).
  - **Skill demonstrated:** AI agents, agent orchestration, workflow execution.

**D-30 — out\_of\_scope and vague skip the documentalist**

- **Date:** 2026-10-04 · **Milestone:** v1.10.0 · **User decision, measured before/after**
- **Need:** avoid paying for a document search on requests that do not need one. For out\_of\_scope (another product) and vague (undetermined service), the technician has **no tool** (D-17) and must not assert anything. The passages found by the documentalist were useless at best and misleading at worst: unrelated vector neighbors.
- **Options:** (a) stay on the common path (documentalist then technician); (b) **conditional edge triage → technician without tools**; (c) a **fixed** answer generated by the code, without an LLM ("Je ne couvre que PostgreSQL, nginx et Redis..." — "I only cover PostgreSQL, nginx and Redis...", "Quel service ?" — "Which service?").
- **Choice:** (b). after\_triage sends to the documentalist only the categories that use the knowledge base (NEEDS\_RETRIEVAL = {malfunction, documentation}). history, out\_of\_scope and vague go straight to the technician. An outage with no known service, reclassified as vague by the code, also takes this shortcut.
- **Why (b) rather than (c):** (c) would be even faster (0 LLM calls), but a fixed text adapts neither to the language of the question nor to its content. The technician names the out-of-scope product ("je n'ai pas d'informations sur Kafka" — "I have no information about Kafka") and asks a suitable clarifying question ("quel service, quel symptôme ?" — "which service, which symptom?"). (b) changes only one graph edge and keeps the shared post-processing and the same checks. It is the simplest change that remains correct.
- **Measurements** (same conditions before/after, 5 runs per case and per model, warm; docs/BENCH.md § v1.10.0):

| Model      | Category                | Pass rate | p50 latency          | LLM calls | Tokens in   |
|------------|-------------------------|-----------|----------------------|-----------|-------------|
| Qwen3-4B   | out_of_scope (Kafka)    | 5/5 → 5/5 | 1.26 → 0.82 s (−35%) | 3 → 2     | 1,139 → 724 |
| Qwen3-4B   | vague ("ça marche pas") | 5/5 → 5/5 | 3.07 → 0.44 s (−86%) | 4 → 2     | 2,150 → 710 |
| Qwen2.5-7B | out_of_scope (Kafka)    | 5/5 → 5/5 | 1.96 → 2.02 s (+3%)  | 3 → 2     | 1,136 → 724 |
| Qwen2.5-7B | vague ("ça marche pas") | 5/5 → 5/5 | 3.87 → 1.74 s (−55%) | 4 → 2     | 1,968 → 710 |

Full benchmark (6 cases × 3, each model): 4B 15/18 → 15/18, 7B 18/18 → 17/18. The 7B gap comes from C6 (malfunction path, **unchanged**). Over 10 additional runs, old and new code both score 8/10, so the gap is not a regression of the routing change. Same for C3 with the 4B (old 5/8, new 7/8).

**Correction (v1.11.2, review F-3):** this account first attributed both 7B failures on C6 to wording only. In the 10-run control on the new code (docs/bench/bench-20261004-124927.md) the two failures are different: - run 1, check "says the check failed": the wording "statut indéterminé... limitation de l'outil" ("undetermined status... tool limitation") was not recognized by the keyword check; - run 3, check "no status asserted": the 7B wrote "il semble que le serveur Redis soit en cours d'exécution mais refuse les connexions" ("the Redis server seems to be running but refuses connections"), an **invented observation** after a tool failure.

The old check flagged run 3 for the wrong sentence (a suggested check, "vérifiez que le service Redis est en cours d'exécution") and would have missed the real, hedged claim ("soit"). v1.11.2 fixes asserts\_status in cases.py: suggested checks ("vérifiez que", "assurez-vous que", "make sure", "check that", "ensure", "verify that") are no longer counted as claims, and hedged claims ("soit", "semble être", "seems to be", "appears to be") are (tests/test\_cases\_postprocess.py::test\_status\_assertion\_ignores\_suggested\_checks\_and\_catches\_hedged\_claims). Re-scoring every archived run with the fixed check changes only 2 rows: the two 7B C6 failures of the full v1.11.2

run (bench-20261004-190049.md), both "assurez-vous que le service est en cours d'exécution", i.e. false positives; the 7B moves from 16/18 raw to 18/18 re-scored. A new 10-run 7B C6 control on v1.11.2 with the fixed check (bench-20261004-190503.md) scores 10/10. In short: the 7B did invent a status once in the v1.10.0 control (1/10), and did not in the 13 v1.11.2 runs. The 4B scores 1/3 on C6 in the full v1.11.2 run (bench-20261004-185627.md: 2× "says the check failed", 1× "exact citations"). - **Trade-offs / findings:** - For out\_of\_scope with the 7B, **no latency gain**. The old documentalist already did not search (1 short LLM call), and the technician's answer (~100 tokens) dominates. The gain is then limited to input tokens (-36%). - The technician no longer sees a "brief" for these categories. It did not need one: it receives the question, the category and an explicit instruction. - A triage that wrongly classifies a real outage as vague now also loses the search. The risk already existed (no status tool), and triage scores 12/12 on the control set (D-24, measured ad hoc, raw output not archived). - **Rollback:** tag pre-1.10.0 (v1.9.3); the change is contained in after\_triage. - **Skill demonstrated:** orchestration, latency/cost, evaluate → engineering plan.

## D-31 — Web demo: the real pipeline, a live trace, one question at a time

- **Date:** 2026-10-04 · **Milestone:** v1.11.0
- **Need:** show the system in a browser, which a terminal cannot do. It must show the triage category, the path through the graph, each agent's loop, the SQL written by the model and its rows, the guardrails, all **without duplicating the logic**.
- **Options:** (a) FastAPI + WebSocket; (b) **Starlette + Server-Sent Events** and a framework-free HTML/JS page; (c) Gradio or Streamlit.
- **Choice:** (b). webapp.py calls workflow.run\_request and relays to the browser, over SSE, each trace event already emitted by the graph (on\_event). The page (static/) only displays them.
- **Why:**
  - SSE is enough, because the stream flows in one direction only (server → browser); it is native in the browser (EventSource) and easy to test;
  - Starlette and uvicorn are already MCP dependencies: no new heavy library;
  - Gradio or Streamlit would have imposed their layout and hidden the trace, which is precisely the point of the demo.
- **Robustness choices:**
  - **one question at a time:** an asyncio lock; subsequent questions wait in a queue and see their position. A single 8 GB GPU and numParallelSessions = 1 on the LM Studio side would make concurrency illusory anyway;
  - **a single warm MCP server**, shared: the RAG models load once (~30 s) at startup, not on every question;
  - **scenario switched warm** via HS\_SCENARIO\_FILE, a small extension point of data\_store tested on its own: restarting the tool server on each switch would have cost ~30 s;
  - **clear errors** (LM Studio unreachable, model missing), a **maximum time** per question, validated parameters (length, model and scenario on an allowlist).
- **HTTP API** (webapp.create\_app):

| Route                                                                             | Returns                                                                                                                                                 |
|-----------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|
| GET /                                                                             | the demo page (static/index.html)                                                                                                                       |
| GET /api/health                                                                   | JSON: version, tool server state (starting, loading, ready, error), busy, waiting, LM Studio status (reachable, available and loaded models), model ids |
| GET /api/config                                                                   | JSON: models, scenarios, the six cases, docs_url                                                                                                        |
| GET /api/ask?<br>question=&model=s1m\ large&scenario=stopped\ running\ tool_error | a Server-Sent Events stream: accepted, queued (position), status, trace (each graph event), result, error, done                                         |

The question must be 1–500 characters (otherwise HTTP 400, as for an unknown model or scenario). A keep-alive comment is sent every 15 s. Each web question is saved under runs/ like a CLI run. Environment variables:

HS\_WEB\_TIMEOUT\_S (per-question timeout, 240 s), HS\_WEB\_QUEUE\_TIMEOUT\_S (maximum wait in the queue, 300 s), HS\_DOCS\_URL (documentation links on the page). Since v1.11.2, the path in the result lists "post-processing" only when the technician produced an answer. - **Trade-offs:** - no authentication: the demo is intended for a local network, and 127.0.0.1 is the default address; - the scenario switch goes through a shared file; this is correct thanks to the lock, but it would be global state to replace with a request parameter in a multi-user version; - a running demo keeps its warm MCP tool server, and therefore the retrieval models (~1 GB, D-10), on the GPU: stop it before running a benchmark on an 8 GB card. - **Verification:** 7 tests (tests/test\_web.py, scripted LLM and in-memory MCP server: SSE stream, SQL rows, path, scenario switch, LM Studio absent, missing model, queue). In Edge, **all 6 cases × 2 models pass** (category, path, trace, answer, sources), on desktop and on mobile. - **Skill demonstrated:** production-grade software, observability, AI agents (shown live).